Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-B Congestive Heart Failure

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-02-C01				
NPLEX-P1-CARD-18-02-C02				
NPLEX-P1-CARD-18-02-C03				
NPLEX-P1-CARD-18-02-C04				
NPLEX-P1-CARD-18-02-C05				
NPLEX-P1-CARD-18-02-C06				
NPLEX-P1-CARD-18-02-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-02-C01-BD1

From a failing left ventricle to alveolar flooding

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C01 Activity: NPLEX-P1-CARD-18-02-C01-BD1 Date: _____

Without looking at your notes, draw the route that pressure takes from a left ventricle that cannot empty or fill normally to fluid inside the alveoli. Name every chamber and vessel in order and write beside each one what happens to its pressure, and mark the place where the absence of valves lets pressure pass backward. Write the normal pulmonary capillary pressure and the normal plasma oncotic pressure with their units, the normal range for pulmonary capillary wedge pressure, and the point at which filtration outruns lymphatic clearance. Then list the causes of left-sided failure in groups: loss of muscle, pressure load, volume load, and direct muscle damage. Finish by naming the two lesions that produce every pulmonary finding while the left ventricle itself stays normal or underfilled.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-02-C01-BD2

Explain three breathing patterns and one blood test

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C01 Activity: NPLEX-P1-CARD-18-02-C01-BD2 Date: _____

A patient with left-sided failure is breathless walking to the bathroom, has to sleep on three pillows, and wakes gasping about two hours after falling asleep. For each of those three findings write the mechanism in two or three sentences, naming what moves where and which receptors drive the breathing pattern. Explain why the breathlessness on lying flat starts within a minute or two while the one that wakes the patient takes one to a few hours. Then write why this patient wheezes and why the crackles are heard at the bases first. Finish with the blood findings of forward failure: write what happens to the BUN to creatinine ratio, give the value that marks the change, and explain the tubular handling of urea that produces it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-02-C01-RT01

Rebuild the seven-step congestion sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C01 Activity: NPLEX-P1-CARD-18-02-C01-RT01 Date: _____

Rebuild the seven-step sequence that runs from a rise in left ventricular end-diastolic pressure to pulmonary edema, one step per line and in order. Beside the step where interstitial fluid appears, write which two forces are now out of balance and which drainage route has been exceeded. Beside the step where lying flat matters, write how much blood shifts into the chest and why a failing ventricle turns that shift into higher capillary pressure rather than more stroke volume. Then write what two things a long history of mitral stenosis changes about where the alveolar flooding step sits on this sequence, and why a patient one day after a large infarct floods at a lower pressure.

Rebuild from memory: From a failing left ventricle to orthopnea and PND

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-02-C02-BD1

Right-sided failure from causes to organs

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C02 Activity: NPLEX-P1-CARD-18-02-C02-BD1 Date: _____

Write out right-sided heart failure from memory in four blocks. First, list the causes in order, starting with the most common one, and separate the ones that arise in the lungs or pulmonary vessels from the ones that do not, naming the acute and the chronic version of each. Second, write how the neck veins are used as a manometer: the vein used, the patient position in degrees, the normal mean right atrial pressure, the height above the sternal angle that counts as raised, and what a sustained response to abdominal pressure means and how long that pressure is held. Third, write the two requirements for dependent edema, why weight gain comes before visible swelling, where the fluid settles in a walking patient and a bedbound one, and why it pits. Fourth, draw the hepatic acinus and label the zone that atrophies and the zone that becomes fatty, then write what the liver looks like grossly, what low output added to congestion produces, and what years of congestion produce.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-02-C02-BD2

Three patients with a raised JVP

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C02 Activity: NPLEX-P1-CARD-18-02-C02-BD2 Date: _____

Three patients each have a jugular venous pulsation 9 cm above the sternal angle. Patient 1 has basal crackles, an S3, and a displaced apex beat. Patient 2 has clear lungs, long-standing chronic obstructive pulmonary disease, and a loud pulmonary component of the second heart sound. Patient 3 has clear lungs, ST elevation in leads II, III, and aVF, and a blood pressure of 82/56 mm Hg. For each patient write which ventricle is failing, where the pressure load or muscle loss sits, and whether the lungs are congested, and explain the lung finding from the position of the lesion in the two circulations. For patient 2 write the pulmonary mechanism that raised the pressure and what the arterioles have done over time. For patient 3 write what has happened to left ventricular filling and give the two reasons, then explain why giving a large volume load helps this patient while it harms patient 1.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-02-C03-BD1

Two failing ventricles, two sounds, one peptide

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C03 Activity: NPLEX-P1-CARD-18-02-C03-BD1 Date: _____

Write the formula for ejection fraction and its normal range, then the three ejection fraction categories with their cutoffs. For the reduced type, write what has happened to the end-systolic pressure-volume relationship, which direction sarcomeres are added, and use the law of Laplace, stated as a formula, to explain the wall stress, the oxygen demand, and the leaking valve that follow. For the preserved type, write what active relaxation needs and the pump it uses, then list what raises passive stiffness, what happens to the end-diastolic pressure-volume relationship, and which patients have it. Next, describe the S3 and the S4: the timing in diastole, the pitch, where each is heard, what is happening in the ventricle at that moment, which failure type each accompanies, when an S3 is normal, and which rhythm abolishes the S4. Finish with BNP: write the precursor and the two fragments, which receptor and second messenger the active one uses, how each fragment is cleared, the rule-out values for both, and four states that raise the level and one that lowers it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-02-C03-BD2

Why the numbers can look reassuring

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C03 Activity: NPLEX-P1-CARD-18-02-C03-BD2 Date: _____

A woman of 78 with long-standing hypertension and a body mass index of 38 arrives with pulmonary edema that began when her heart rate rose to 150 per minute in atrial fibrillation. Her ejection fraction is 62%, her chest radiograph shows a cardiothoracic ratio of 0.46, and her BNP is modestly raised. Write which failure type this is and defend it from the history and the geometry of her ventricle. Then explain, one at a time, why a normal ejection fraction does not exclude failure here, why her BNP is lower than you would expect for her filling pressure and name both the wall stress reason and the body habitus reason, and why her chest radiograph shows no enlarged heart. Next write why the arrhythmia alone tipped her into edema, naming the two things a fast atrial fibrillation takes away from a stiff ventricle. Finish by writing what her radiograph would show in order as her wedge pressure climbed from normal to above 25 mmHg, with the pressure band beside each sign, and write why those signs would appear late if her failure had been building for years.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-02-C03-CP01

Reduced against preserved ejection fraction

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	HFrEF (systolic)	HFpEF (diastolic)
Ejection fraction cutoff		
Core defect in the muscle		
Pressure-volume loop change		
Ventricular size, wall thickness, and direction of sarcomere addition		
Typical causes		
Extra heart sound and what produces it		
How much the ventricle depends on atrial contraction		
Why the filling pressure rises in each		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-02-C04-BD1

Lay out the three compensatory systems

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C04 Activity: NPLEX-P1-CARD-18-02-C04-BD1 Date: _____

Write what the baroreceptors and kidney are actually sensing when they switch on their defenses in heart failure, and define the volume term that explains why they stay switched on while total extracellular volume is expanded. Then take the three systems one at a time. For the sympathetic nervous system, write what reduced baroreceptor firing does to outflow, then list the beta-1, alpha-1, and venous effects, and the harms that appear over months including what happens to the receptors themselves and the proteins that do it. For the renin-angiotensin-aldosterone system, write the three stimuli for renin release, then six actions of angiotensin II at its receptor, then what aldosterone does, in which cell, and at the expense of which two ions. For antidiuretic hormone, write which stimulus is driving it here, the receptor, the channel it inserts, and the electrolyte result. Finish by writing why retained volume adds little stroke volume in this setting, naming the curve and the part of it the ventricle is sitting on.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-02-C04-BD2

Why blocking hormones works and squeezing harder does not

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C04 Activity: NPLEX-P1-CARD-18-02-C04-BD2 Date: _____

Two patients with reduced ejection fraction are treated along different lines. One has beta-1 receptors, angiotensin II, and aldosterone blocked, and over months the ventricle becomes smaller and less spherical. The other is given a drug that raises contractility alone, and the ventricle keeps enlarging. Write what remodeling is, listing the changes in the myocytes, the genes they re-express, the cells that die and how, and the hormones that drive the fibrosis. Then explain from those mechanisms why the first course changes the course of the disease and the second does not, and write what a larger more spherical ventricle does to wall stress and to the papillary muscles, and why that makes the process self-sustaining. Next write where the natriuretic peptides fit: which chambers release which peptide, the receptor and second messenger they work through, the three actions that oppose the other systems, and the three reasons they cannot win. Finish by predicting the serum sodium and the urine sodium of a patient with advanced disease and marked edema, with the mechanism for each in one line.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-02-C04-RT01

Rebuild the neurohormonal cycle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C04 Activity: NPLEX-P1-CARD-18-02-C04-RT01 Date: _____

Draw the neurohormonal cycle of heart failure as a closed loop with seven steps, numbering each step and writing it out in words so the loop returns to its starting point. Mark on the loop the one step that is genuinely useful in a short crisis and the two steps where the response begins to harm the heart. Beside the step on vasoconstriction and retention, write which variable of ventricular work each one raises. Beside the remodeling step, write the three hormones responsible and the tissue change each one causes. Then write the single sentence that explains why a response built for hemorrhage damages a failing pump, and write where on your loop a drug that blocks an angiotensin II receptor and a drug that blocks a mineralocorticoid receptor each interrupt it.

Rebuild from memory: The neurohormonal cycle of heart failure

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-02-C04-PR01

Predict what six months of neurohormonal drive does

A man with an ejection fraction of 25% after an infarction has had six months of untreated sympathetic, renin-angiotensin-aldosterone, and antidiuretic hormone activation. Predict each item below, write the direction of the change, and give the hormone, receptor, or mechanism that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to systemic vascular resistance and so to the work the ventricle does against it		
What happens to preload and to the venous pressure behind both ventricles		
What happens to the number of beta-1 receptors on the myocyte surface and how		
What happens to serum potassium and to serum sodium		
What happens to the interstitial collagen content of the myocardium		
What happens to ventricular size and shape, and what that does to wall stress		
What happens to the ability of natriuretic peptides to offset all of the above		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-02-C05-BD1

Organise the complications by their cause

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C05 Activity: NPLEX-P1-CARD-18-02-C05-BD1 Date: _____

Write the three headings that complications of heart failure fall under, then place each complication below under its heading and write its mechanism. Cover acute pulmonary edema with its five triggers and what the sympathetic surge adds, cardiogenic shock with its three numerical criteria and the fraction of ventricle classically lost, the atrial and the ventricular arrhythmias with the membrane changes behind each, mural thrombi with the three parts of the triad named and the sites they form and travel to, cardiorenal syndrome with both the arterial and the venous mechanism, and cardiac cachexia with its three contributors. Then write the three-number haemodynamic profile that separates cardiogenic shock from hypovolaemic shock and from early septic shock, using output, wedge pressure, and systemic vascular resistance for all three.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-02-C05-BD2

Four deteriorations in one patient

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C05 Activity: NPLEX-P1-CARD-18-02-C05-BD2 Date: _____

A man with an ejection fraction of 20% deteriorates on four separate occasions. On the first he develops severe breathlessness with frothy pink sputum and diffuse crackles over two hours. On the second he is found to have a wedge pressure of 10 mmHg with the same lung picture. On the third his creatinine rises while his jugular venous pressure is markedly raised and his blood pressure is 96/60 mm Hg. On the fourth he has a sudden right-sided weakness. For each occasion write what has happened and the mechanism, and for the second write what the normal wedge pressure tells you about where the fluid is coming from. For the third, write the two separate mechanisms by which his kidney is being injured and explain how a raised venous pressure lowers filtration using the pressures inside the capsule. For the fourth, write where the material formed and why, naming the three conditions that allowed it and the appearance it would show on section. Finish by writing why his weight has not risen despite progressive muscle loss.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-02-C05-CA01

Cold, wet, and oliguric three days after an infarction

A 64-year-old man is in hospital three days after an anterior myocardial infarction that was not reperfused. Over six hours he becomes confused and breathless. Temperature is 35.8 C (96.4 F), pulse is 118/min, respirations are 30/min, and blood pressure is 84/58 mm Hg. His hands and feet are cold and mottled, his skin is clammy, the jugular venous pulsation is 7 cm above the sternal angle, and there are crackles over both lower and mid zones with an S3 at the apex. Urine output has been 12 mL per hour for four hours. A pulmonary artery catheter shows a cardiac index of 1.8 L/min/m2, a pulmonary capillary wedge pressure of 26 mmHg, and a systemic vascular resistance well above normal. Mixed venous oxygen saturation is 44%, serum lactate is raised, creatinine has risen from 94 to 168 micromol/L, and serum sodium is 129 mmol/L.

1. Recognize. Name the complication of heart failure this patient has developed and state which of its numerical criteria his measurements meet.

2. Explain. Explain the mechanism of the cold mottled skin, the crackles with the wedge pressure of 26 mmHg, the mixed venous saturation of 44%, and the raised lactate, taking each one back to either low forward output or high filling pressure.

3. Differentiate. Write how his three measured numbers would differ in hypovolaemic shock and in early septic shock, using cardiac output, wedge pressure, and systemic vascular resistance for each.

4. Investigate. Write which further findings you would look for to explain the falling urine output and the serum sodium of 129 mmol/L, and what each finding would show.

5. Decide. State which single mechanism makes this state self-reinforcing once the blood pressure has fallen, and name the organ whose perfusion depends on that pressure.

6. Justify. Justify that answer from coronary perfusion and from the proportion of left ventricle that must be lost for this complication to appear, and explain why the sympathetic surge that accompanies it adds to the problem rather than solving it.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-02-C06-BD1

How a high output still leaves a congested patient

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C06 Activity: NPLEX-P1-CARD-18-02-C06-BD1 Date: _____

Define high-output heart failure and name its defining haemodynamic feature. Write the relationship between mean arterial pressure, cardiac output, and systemic vascular resistance as a formula, then use it to explain in four or five steps why a fall in resistance ends in congestion and edema, naming the receptors and hormones that are activated and what they retain. Then work the fistula numbers from memory: starting from a mean arterial pressure of about 93 mmHg and an output of 5 L/min, calculate the resistance in Wood units, show what the resistance falls to when a large fistula is added, and calculate the output needed to hold the pressure near 90 mmHg. Write why the tissues gain little from that extra flow, what the venous blood looks like, and what happens to the heart rate when the fistula is compressed and why. Finish by listing five causes of high-output failure with the mechanism for each, and for the thiamine one name the three enzymes that lose their cofactor.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-02-C06-BD2

Sort two congested patients by their resistance

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C06 Activity: NPLEX-P1-CARD-18-02-C06-BD2 Date: _____

Two patients have orthopnea, raised jugular venous pressure, ankle edema, and an S3. Patient 1 has cool pale hands, a weak pulse with a narrow pulse pressure, and a mixed venous oxygen saturation of 42%. Patient 2 has warm flushed hands, a bounding pulse with a wide pulse pressure, and a mixed venous oxygen saturation of 78%. Write which has the low output and which the high output, then explain each of the three paired findings from the resistance, naming what the arteriovenous oxygen difference is doing in each patient and why. Write why both patients have the same congestive findings even though their outputs are opposite. Then, for patient 2, write the questions you would ask and the findings you would look for to separate severe anemia, thyrotoxicosis, an arteriovenous fistula, wet beriberi, and extensive Paget disease, giving one discriminating clue for each. Finish by explaining why a patient whose heart was already diseased reaches congestion sooner than a patient whose heart was normal when the resistance fell.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-02-C07-BD1

Rupture to shock, and why the murmur can be quiet

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C07 Activity: NPLEX-P1-CARD-18-02-C07-BD1 Date: _____

Draw the mitral apparatus and label the leaflets, the chordae, and both papillary muscles with the arteries that supply each. Write the causes of chordal rupture and the causes of papillary muscle rupture, and for the infarct-related cause give the number of days after the infarct, the reason the tissue is weakest then, which papillary muscle ruptures and with which infarct, and why that muscle is the vulnerable one. Then rebuild the six-step sequence from rupture to cardiogenic shock in order, and beside the atrial step write what the pressure tracing shows and what makes that atrium unable to absorb the volume. Write why the murmur may be early systolic, soft, or absent, and why the heart looks normal in size on the radiograph. Finish by writing the two findings that separate ventricular septal rupture from this lesion.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-02-C07-BD2

Work the volumes behind a normal ejection fraction

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-B Congestive Heart Failure Competency: NPLEX-P1-CARD-18-02-C07 Activity: NPLEX-P1-CARD-18-02-C07-BD2 Date: _____

A patient with acute mitral regurgitation is in shock with an ejection fraction of 75%. Starting from end-diastolic volume 120 mL and end-systolic volume 30 mL, calculate total stroke volume, ejection fraction, forward stroke volume if 50 mL enters the left atrium each beat, the regurgitant fraction, and forward cardiac output at a heart rate of 100 per minute. Show your working and define regurgitant fraction in one line. Then write why the ejection fraction rose after the rupture even though the patient is in shock, and explain what the ejection fraction is measuring that forward output is not. Next write where those 50 mL per beat go and what they do to the wedge pressure. Finish by writing what the same regurgitant volume would do in a patient who had carried mitral regurgitation for fifteen years, naming the two structural adaptations that patient has and this one lacks, and explain from compliance why one gets sudden pulmonary edema and the other does not.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-02-C02-Q01

Hypotension with a raised JVP and clear lungs

A 59-year-old man comes to the emergency department because of 4 hours of chest pain. Pulse is 54/min and blood pressure is 80/54 mm Hg. The jugular venous pulsation is 10 cm above the sternal angle and rises further when pressure is applied to the abdomen. The lung fields are clear on auscultation and on chest radiograph, and there is no S3. The electrocardiogram shows ST elevation in leads II, III, and aVF. Which of the following best explains the hypotension in this patient?

A. Alveolar flooding from a rapid rise in pulmonary capillary pressure

B. Reduced left ventricular filling because right ventricular output has fallen and the septum has shifted leftward

C. A fall in systemic vascular resistance from arteriolar vasodilation

D. Acute regurgitation of blood into a small, noncompliant left atrium

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-02-C06-Q02

Congestion with warm extremities on dialysis

A 47-year-old woman receiving hemodialysis through a large upper arm arteriovenous fistula comes to the physician because of 3 weeks of breathlessness on lying flat and swelling of both ankles. Temperature is 36.8 C (98.2 F), pulse is 104/min, and blood pressure is 142/58 mm Hg. Her hands are warm, the pulse is bounding, the jugular venous pulsation is raised, and an S3 is present at the apex. Cardiac output is 9 L/min and mixed venous oxygen saturation is 79%. Which of the following best explains her mixed venous oxygen saturation?

A. A large share of the cardiac output returns to the heart without passing through capillaries

B. Tissues are extracting more oxygen per unit of blood delivered

C. Pulmonary edema is limiting oxygen uptake across the alveolar membrane

D. Reduced hemoglobin concentration has lowered the oxygen carrying capacity of blood

Why did you pick it?